



COMSATS University Islamabad, Lahore Campus

Department of Computer Engineering

Assignment 4 – FALL 2025

Course Title:	Digital System Design	Course Code:	CPE 344	Credit Hours:	4(3,1)
Course Instructor:	Dr. Muhammad Naeem Awais	Program Name:	BS Computer Engineering		
Semester:	7 th	Batch:	FA22	Section:	A, B
Date Given:	16 th December, 2025				
Submission Date:	18 th December, 2025		Maximum Marks:	20	
Name:			Registration Number:		

Important Instructions / Guidelines:

- Do your own work, PLAGARISM will be graded as ZERO
- No late submission.

Question 1:

[CLO2-PLO3-C5]

[20 Marks]

- a. Design and implement a function $F(a,b,c,d) = ab' + cd'$ using any number of 3-input 2-output CLBs. The CLBs are interconnected through programmable connection using switch matrix.
- b. For the part (a) design the bit-file necessary to program the Field Programmable Gate Array (FPGA) using the switching matrix and configurable logic block.